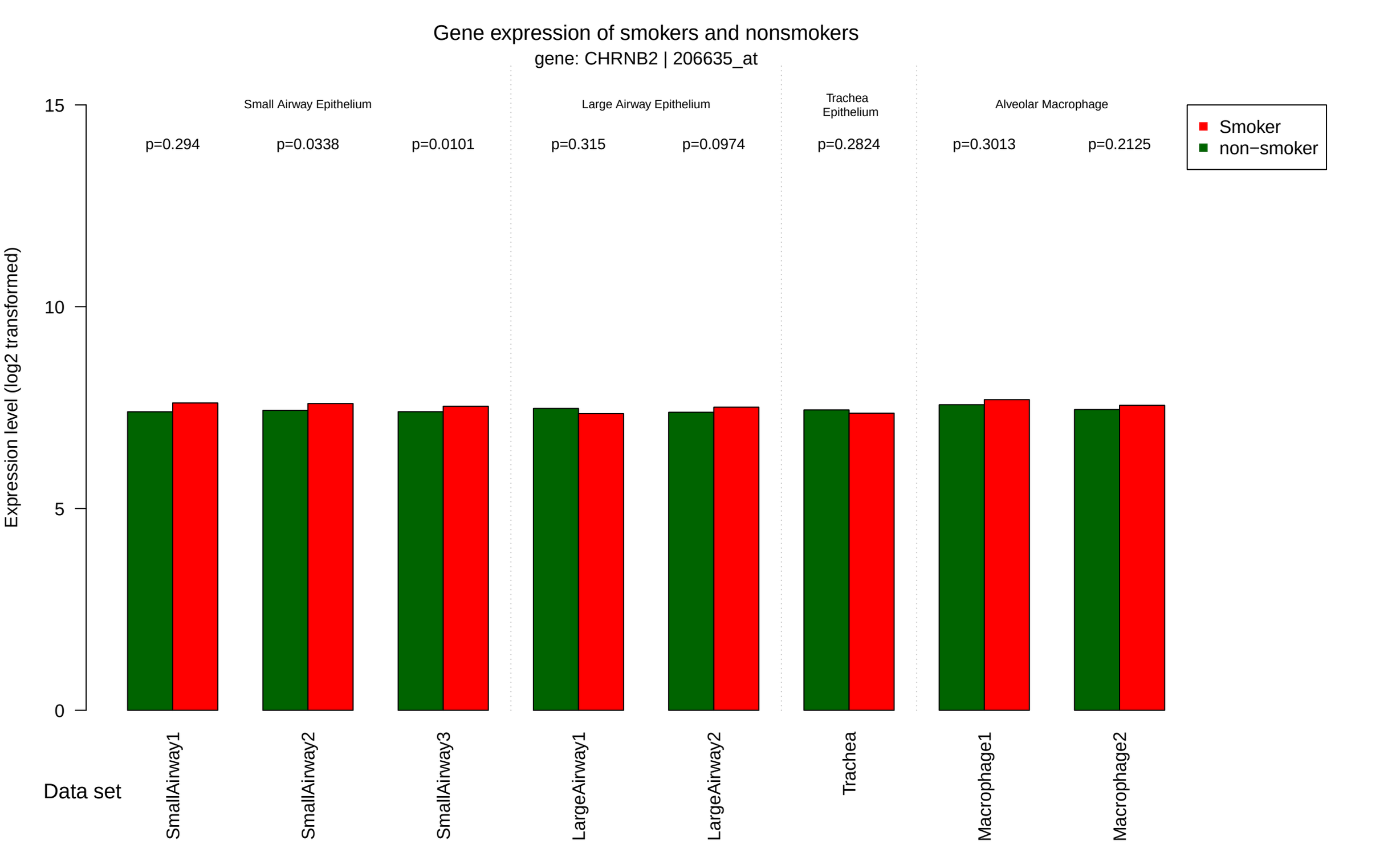
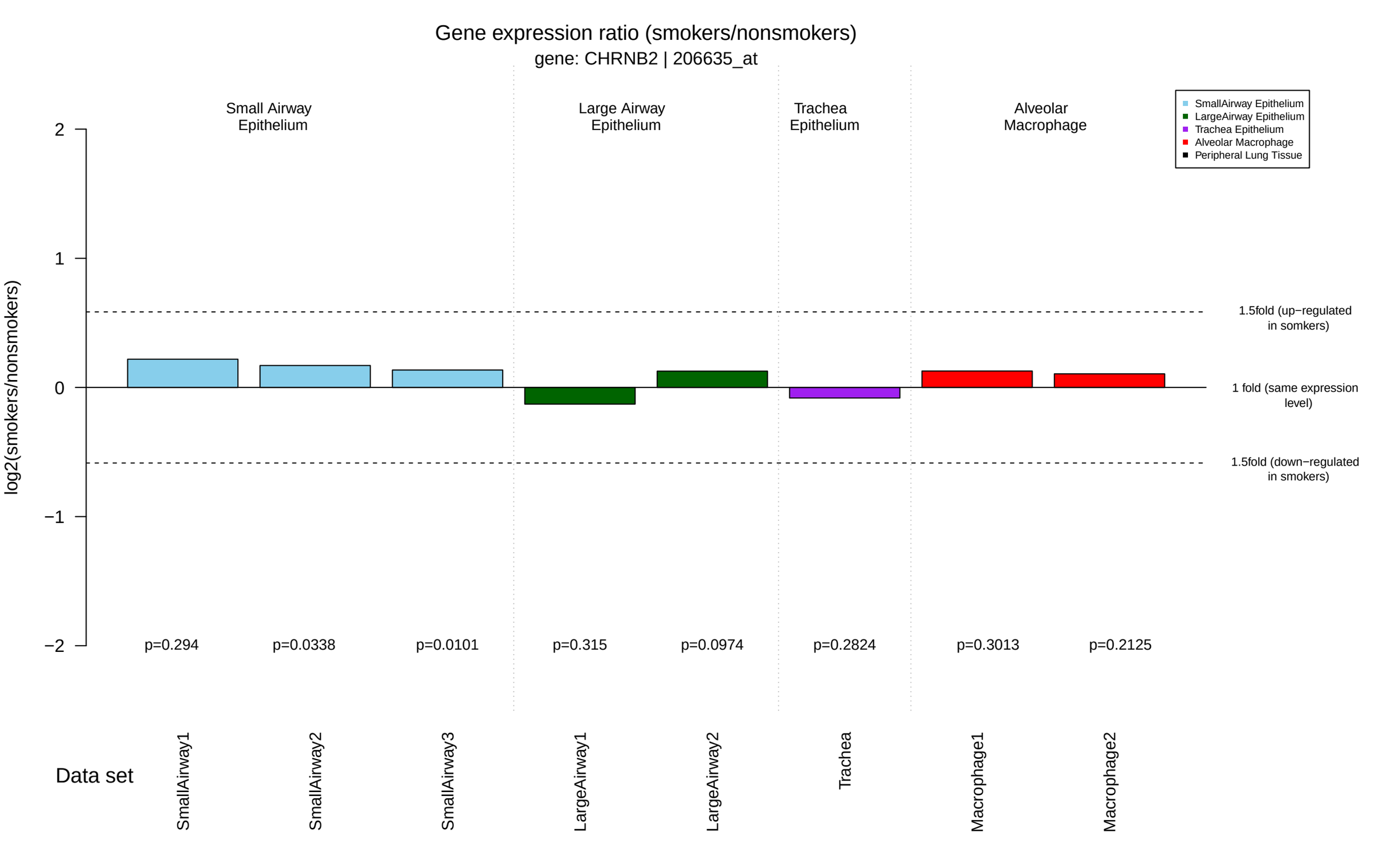
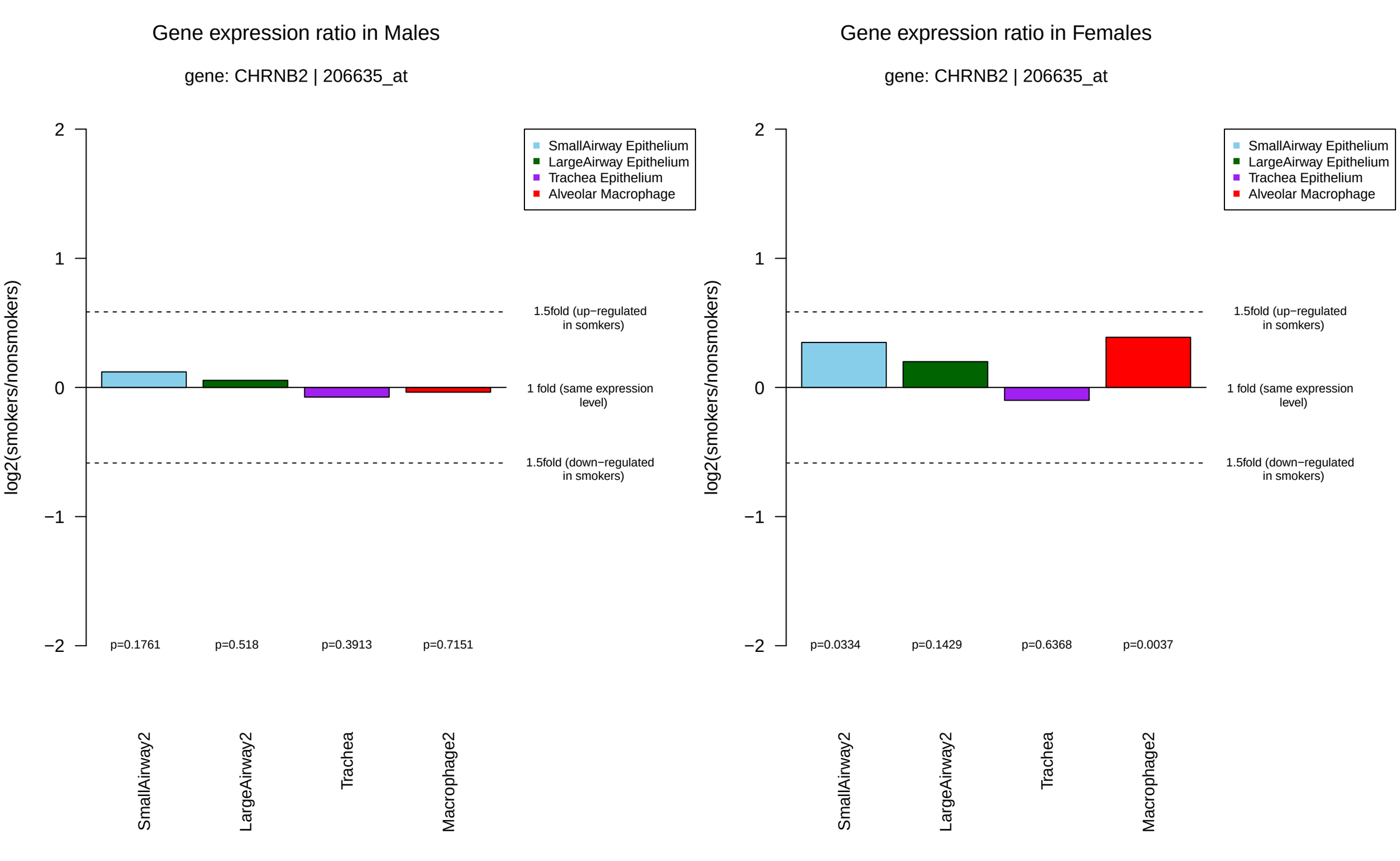
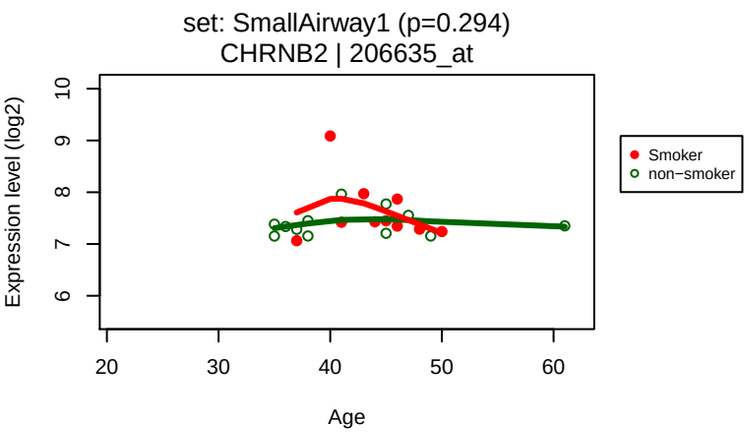
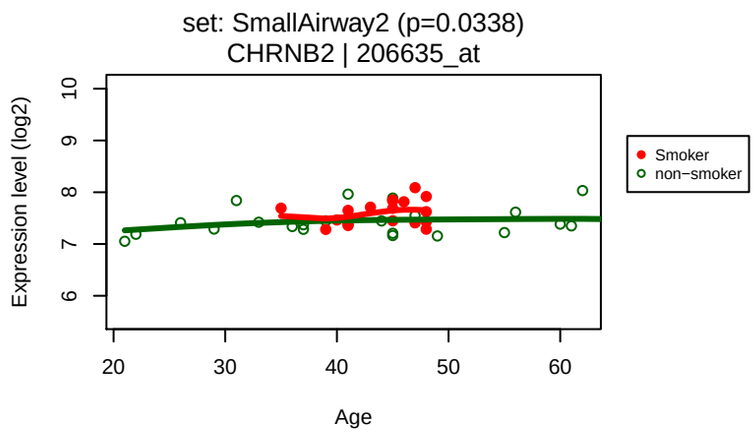
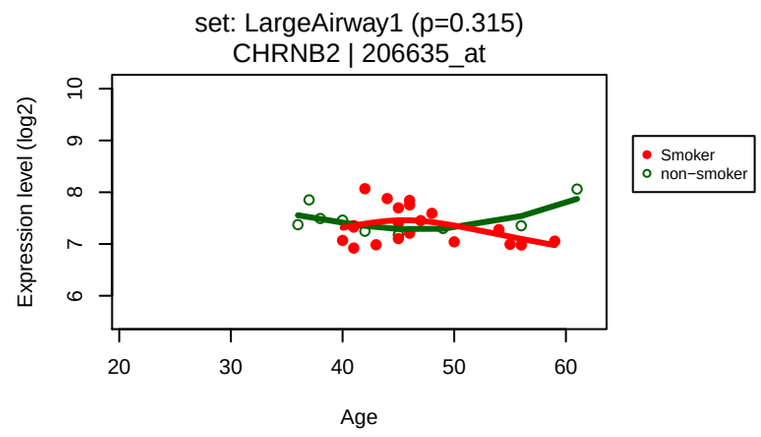

Gene expression of smokers and nonsmokers
gene: CHRNB2 | 206635_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.294
p=0.0338
p=0.0101
p=0.315
p=0.0974
p=0.2824
p=0.3013
p=0.2125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

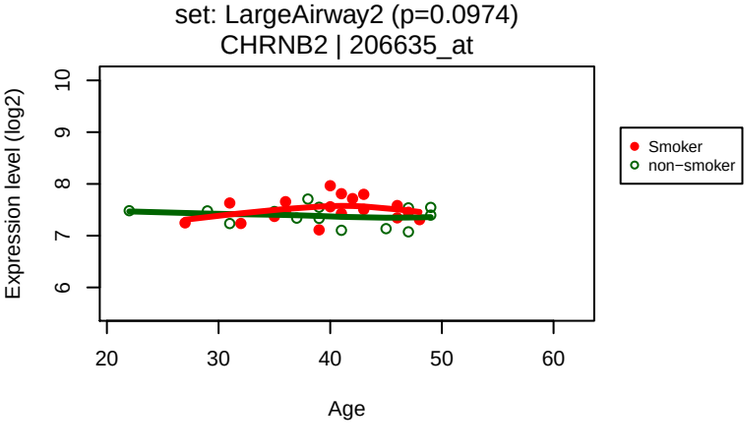

Gene expression ratio (smokers/nonsmokers)
gene: CHRNB2 | 206635_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.294
p=0.0338
p=0.0101
p=0.315
p=0.0974
p=0.2824
p=0.3013
p=0.2125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

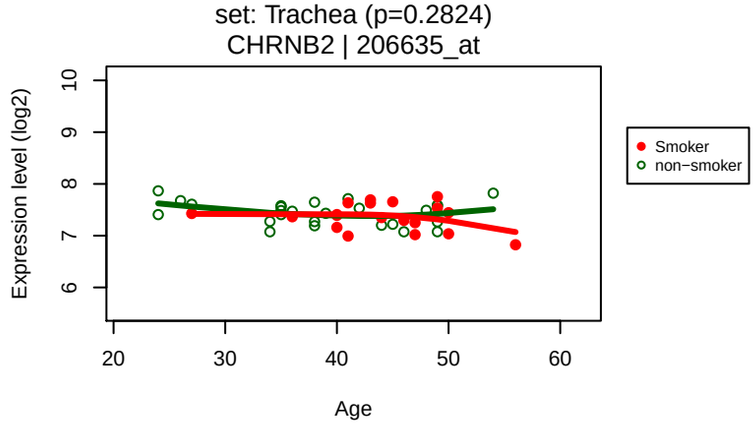

Gene expression ratio in Males
gene: CHRNB2 | 206635_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1761
p=0.518
p=0.3913
p=0.7151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNB2 | 206635_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0334
p=0.1429
p=0.6368
p=0.0037
SmallAirway2
LargeAirway2
Trachea
Macrophage2

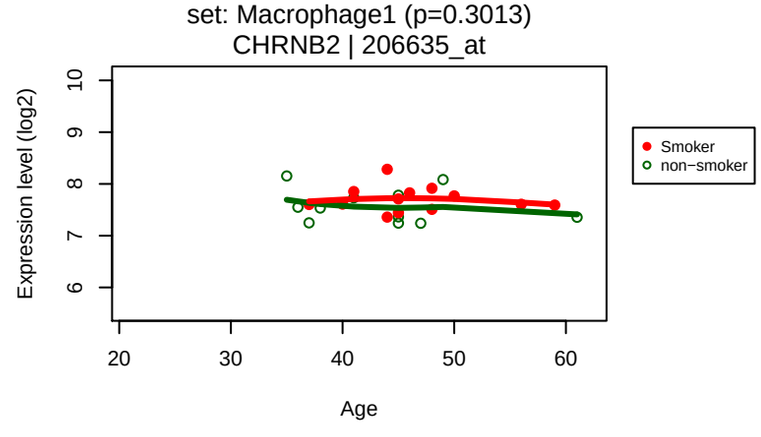

set: SmallAirway1 (p=0.294)
CHRNB2 | 206635_at
Expression level (log2)
Age
Smoker
non−smoker

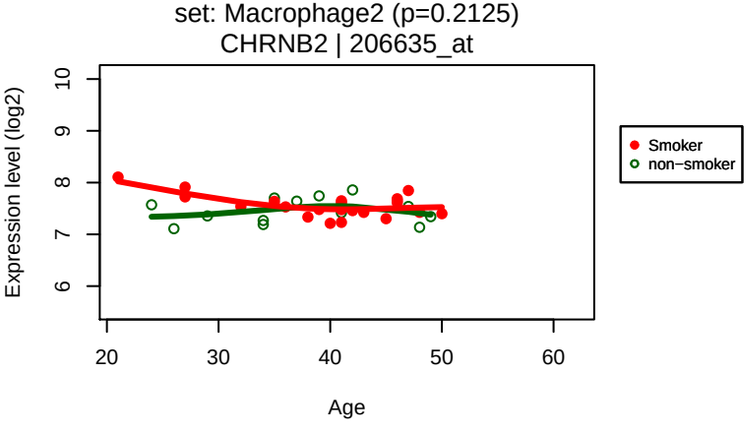

set: SmallAirway2 (p=0.0338)
CHRNB2 | 206635_at
Expression level (log2)
Age
Smoker
non−smoker

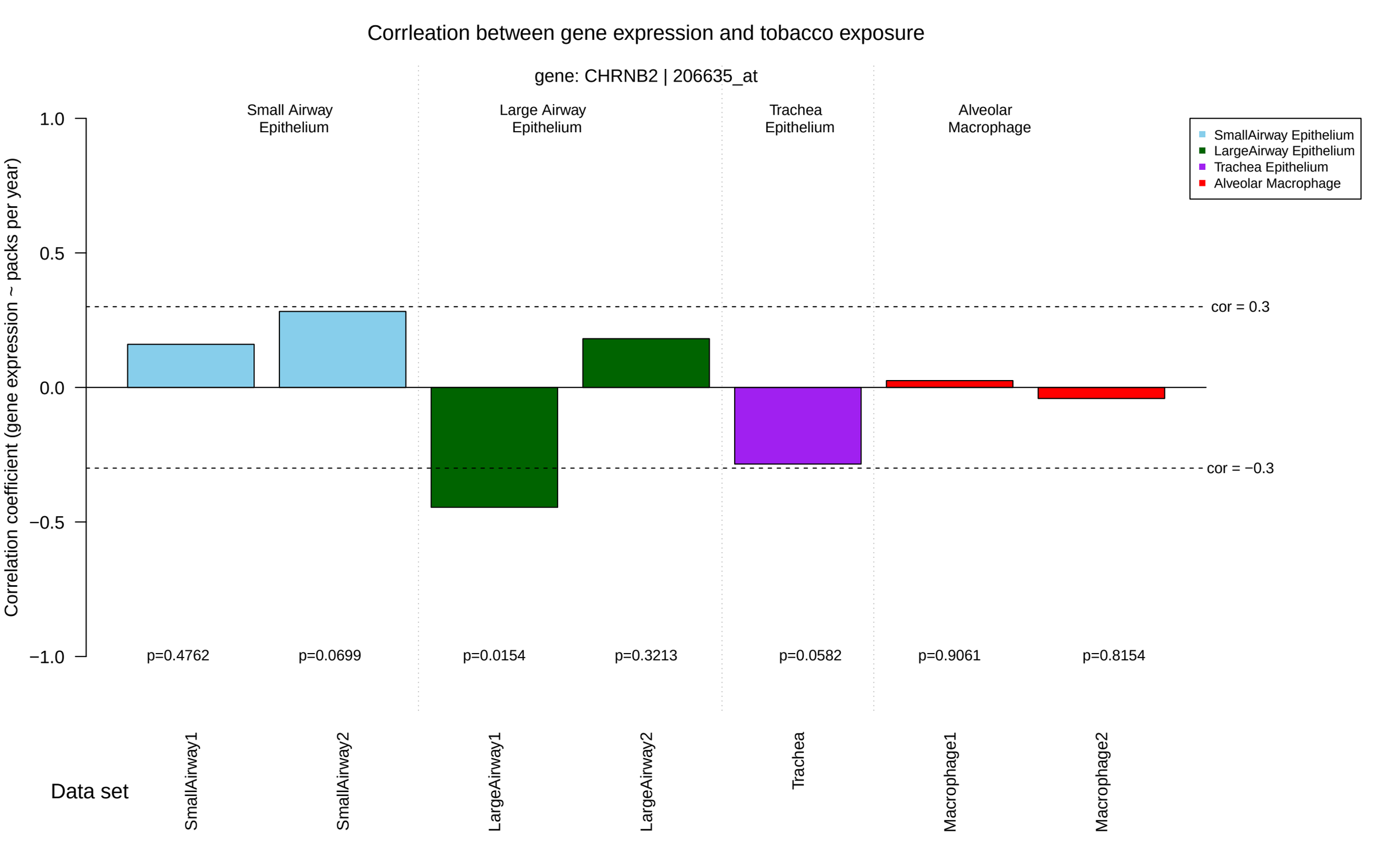

set: LargeAirway1 (p=0.315)
CHRNB2 | 206635_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0974)
CHRNB2 | 206635_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2824)
CHRNB2 | 206635_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3013)
CHRNB2 | 206635_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2125)
CHRNB2 | 206635_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNB2 | 206635_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4762
p=0.0699
p=0.0154
p=0.3213
p=0.0582
p=0.9061
p=0.8154
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2